

Joseph Walton

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Professional Summary

Lead data scientist and machine learning engineer with over 10 years of experience building production-grade AI, machine learning and data systems. Hands-on experience delivering LangGraph-based agents, LLM-powered applications, RAG, semantic search and evaluation frameworks in large government and commercial organisations. Strong Python engineer with a track record of orchestrating complex data and model workflows, integrating multiple data sources, and improving the accuracy, reliability and observability of AI products. Experienced technical leader who turns ambiguous user and business needs into reusable solutions and communicates effectively with technical and non-technical stakeholders.

Technical Skills

- **AI Agents & LLMs:** LangGraph, LangChain, Azure AI Foundry, Azure OpenAI, RAG, agent orchestration, human-in-the-loop workflows
 - **Natural Language Systems:** NLP, Hugging Face, BERT, FastText, semantic search, vector search, recommendation systems, query relevance evaluation
 - **Programming & Data:** Python, SQL, Java, Polars, Pandas, NumPy, Apache Spark, PySpark
 - **Testing & Production AI:** LLM evaluation, observability, red teaming, MLOps, MLflow, CI/CD, Docker
 - **Cloud & Orchestration:** Microsoft Azure, Azure Functions, Azure ML Pipelines, Vertex AI Pipelines, Google Cloud Platform, AWS
 - **Data Platforms:** Elasticsearch, PostgreSQL, MySQL, Cloudera
 - **Machine Learning:** Scikit-learn, TensorFlow, PyTorch
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Professional Experience

IBM

Lead Data Scientist

June 2025 - Present (Contract)

Delivering production AI solutions for central government.

- Designed and delivered a LangGraph-based data processing agent using Python, Azure AI Foundry and Azure Functions, saving each case worker two hours per day and approximately £2M annually.
- Built orchestration and human-in-the-loop processing around complex case data, converting operational requirements into a reliable, production-ready agent workflow.
- Rescued underperforming AI projects by introducing systematic evaluation and observability practices to validate quality and reliability.
- Developed a Claude Skill to streamline human-in-the-loop creation of evaluation datasets for LLM and agent products, saving approximately one hour per evaluation case.
- Re-engineered a large-scale data matching algorithm from Apache Spark to Polars, reducing runtime from two hours to ten minutes and cloud costs by approximately £10k per month.

Tech Stack: Python, LangGraph, Azure AI Foundry, Azure Functions, Polars, Apache Spark

Intellectual Property Office

Senior Machine Learning Engineer

November 2021 - April 2025

Developed and operationalised natural language and generative AI systems for trade mark services.

- Developed LLM-powered applications within a service-oriented architecture using Python, LangChain, Azure OpenAI and Azure Functions.
- Built a RAG agent that interpreted natural-language needs and retrieved relevant Goods and Services terms for novice users.
- Implemented and tuned Elasticsearch HNSW vector search, balancing throughput and recall through systematic evaluation.
- Used relevance metrics and experimental evidence to improve search and recommendation quality by 20%.
- Designed evaluation approaches and conducted red-team testing to identify reliability and adversarial robustness issues in generative AI applications.
- Set the product roadmap for AI applications, translating user needs and delivery constraints into prioritised technical initiatives.
- Implemented an end-to-end MLOps strategy across multiple environments, reducing model deployment time to under one day.
- Architected reusable ETL pipelines in Polars, replacing Azure Databricks workloads, reducing compute costs by 50% and enabling zero-downtime deployments.
- Communicated AI implementation approaches to technical and non-technical stakeholders across international IP offices and contributed to a cross-government AI working group.

Tech Stack: Python, LangChain, Azure OpenAI, Hugging Face, Azure Functions, Azure ML Pipelines, Elasticsearch, Polars

Incopro

Lead Machine Learning Engineer

February 2020 - November 2021

Led delivery of production machine learning systems using large-scale multilingual text and image data.

- Led the design of an MLOps framework that orchestrated training, evaluation, approval and deployment with Vertex AI Pipelines, reducing time to market to under one day.
- Built Spark pipelines to combine and transform large-scale data for model training, batch inference and recommendation workloads.
- Developed and deployed a multilingual BERT model for detecting intellectual property infringements across more than 90 languages.
- Implemented a recommendation engine combining text and image similarity to improve infringement detection.
- Mentored a software engineer through a transition into a machine learning role.

Tech Stack: Python, Hugging Face, TensorFlow, PySpark, Vertex AI Pipelines, Google Cloud Platform, MLflow

Machine Learning Engineer

November 2018 - February 2020

- Designed and built a production NLP system for predicting intellectual property infringement in a secure, containerised environment, increasing enforced infringements by 10% per analyst across approximately 200 analysts.
- Built and deployed an image similarity model that helped secure a major client contract representing 10% of annual recurring revenue.

Tech Stack: Python, TensorFlow, FastText, Google Cloud Platform, MLflow, Docker

Office for National Statistics

Machine Learning Engineer

September 2017 - November 2018

- Developed machine learning models and scalable data pipelines to link and process government administrative data from multiple sources.
- Developed a highly scalable backend for the UK Census 2021 data collection system.

Tech Stack: Python, PySpark, Cloudera

Barclays

Data Scientist (Secondment)

September 2018 - October 2018

- Applied statistical and analytical methods to payments data to create regional economic indicators, continuing the work of the winning ONS x Barclays Hackathon team.
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Purple Secure Systems

Software Engineer

August 2014 - October 2016

- Developed big data processing systems for government and defence clients using Java, Python and PySpark.
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Education

University of Bath

Master of Science in Applied Mathematics

2016 - 2017

Bachelor of Science in Physics

2010 - 2013

Honours & Awards

Winner, ONS x Barclays Data Hackathon

Used payments data to create more granular and timely estimates for Gross Value Added (GVA).